Long-Term Spatio-Temporal Graph Attention Network for Traffic Forecasting

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Abstract

Accurate traffic flow prediction is a critical component of intelligent transportation systems and smart cities, playing an essential role in traffic control, transportation planning, and infrastructure development. Numerous recent research studies highlight the need to enhance prediction accuracy by addressing complex temporal and spatial dependencies. However, due to the complexity of these spatio-temporal patterns, achieving accurate traffic predictions is still a main challenge in long-term scenarios. In this context, we first provide a comprehensive overview of the traffic forecasting to locate where research is going on. Then, we develop a Long-Term Spatio-Temporal Graph Attention Network (LSTGAN) architecture designed to analyze long-term historical data to address the above issue. This architecture encodes several previous time steps and extracts temporal patterns using convolutional layers. These features are then combined with the spatial features captured by a spatial attention module and a graph convolution layer to be processed by a temporal attention decoder responsible for making predictions. Experiments on METR-LA and PEMS-BAY datasets show that our proposed architecture outperforms most existing state-of-the-art baselines.

Keywords:

Traffic Forecasting, GCN, Long-term Dependency, Spatial/Temporal Attention

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1. Introduction

Transportation systems are among the most crucial infrastructures in modern cities, allowing the daily commuting and traveling of millions of people. In addition, transportation systems have become more complex because of rapid urbanization and population growth. Modern transportation systems concern road vehicles, rail transport, and various shared travel modes that have emerged in recent years. This expansion generates many transportation-related problems, such as air pollution and traffic congestion. Early intervention based on traffic forecasting is a key solution for optimizing transportation system efficiency and addressing transportation-related challenges. Thus, traffic flow forecasting helps traffic managers predict and control prospective road congestion in real time. In addition, this system enables travelers to plan or adjust their trips (Wang & Shi, 2013). Therefore, accurate traffic flow forecasting is crucial. To facilitate this, a dashboard for visualizing traffic data has been developed (Moghadas et al., 2024a).

Moreover, traffic flow forecasting is an important task in spatio-temporal forecasting. It exploits the spatio-temporal dependencies of traffic flow data (Guo et al., 2019). Both short-term and long-term traffic forecasting problems are considered in the literature. Practically, challenges in this field are due to the complex spatio-temporal dependencies, short-term and long-term issues (Vlahogianni et al., 2004), dynamic and adaptive aspects, synchronous challenge, multi-scale challenge, and insufficient forecasting precision (Ding et al., 2020; Peng et al., 2020). In addition, time/space/feature dimension interrelations have to be taken into account (Wang et al., 2024e). However, the high construction and maintenance costs limit the widespread application of traffic detectors in transport systems. In many cities, traffic detectors are only sparsely distributed in urban areas, failing to cover the majority of roads, even if taken in several cities and countries in which the traffic is very different (USA (Chen et al., 2001), Colombia (Bernal-Sanchez & Riascos-Ochoa, 2024), China (Li et al., 2024a), Japan (Jiang et al., 2023b), India (Kumar et al., 2024b; Jha et al., 2016), Spain (Lanza et al., 2023), and Belgium (Moghadas et al., 2024a,b)) and thus presenting different chal-

lenges. Moreover, the rapid urbanization in recent years has led to dynamic shifts in traffic patterns and travel demand, providing more challenges for accurate long-term traffic prediction and requiring larger datasets (LargeST (Liu et al., 2023a), XXLTraffic (Yin et al., 2024)).

Many scientific efforts have been reported in the literature to improve graph neural networks in a more progressive way in applications where challenges are becoming more complex. But no algorithm is able to simultaneously address all the key challenges that are present in traffic forecasting during long sequences, as in real cases, because of their numerous features requiring specific methods. In this context, to uncover latent patterns in traffic data, we propose the Long-Term Spatio-Temporal Graph Attention Network (LSTGAN), a model designed to leverage long-term historical data for traffic forecasting. Our architecture is based on an encoder-decoder framework: the encoder captures both temporal trends over extended historical data and spatial correlations among road sensors, effectively integrating these complex dependencies. The decoder further refines this encoded information by selectively focusing on the most relevant temporal and spatial features, enabling accurate traffic predictions. The main contributions of this paper are as follows:

- To effectively capture complex patterns and trends latent in long-term historical traffic data, we propose an encoder with a downsampling process using convolutional layers to reduce the data size while preserving essential information.

- Recognizing that the traffic condition on a particular road is influenced by conditions on adjacent roads, we address spatial dependencies through two modules: one module captures the dynamics of neighboring roads, while the other assesses conditions across the entire road network, allowing for a more holistic understanding of spatial influences.

- We integrate date and time of day information with the previously extracted spatial and temporal dependencies, and we feed this enriched data into a self-attention-based decoder that selectively focuses on the most relevant features for accurate traffic predictions.

- We evaluated our model on two real-world datasets, yielding results that show the superiority of our model over several baseline approaches. An ablation study of our model's components further demonstrated the effectiveness of each module, highlighting their contributions to the overall performance.

The rest of this paper is organized as follows. Section 2 provides a comprehensive and exhaustive overview of existing approaches for traffic forecasting, particularly spatial-temporal graph neural networks (STGNNs), which have been extensively investigated in 2024. Section 3 details the components of the proposed model architecture named Long-Term Spatio-Temporal Graph Attention Network (LSTGAN). Section 4 presents a performance evaluation and an ablation study of the model's components. Finally, Section 5 concludes and outlines future works.

2. A Comprehensive Overview

2.1. From Model-based to Data-based Approaches

Modern traffic flow forecasting with various passive and active sensors began in the 1990s (Larry, 1995; Smith & Demetsky, 1997) and has received very numerous attention since the rise of deep learning and graph-based methods. Indeed, more than 150 papers have been published since 2023, making the last survey (Rahmani et al., 2023) out-of-date and thus justifying an updated, comprehensive, and extensive overview for novices to know where the research is going.

The first methods were model-based, using simulations that were computationally demanding and required tuning model parameters. With the large modern real-time traffic data collection systems, data-driven approaches have become more suitable. Existing approaches can be classified into statistical methods, machine learning methods (Lana et al., 2018; Jiang & Luo, 2022; Rahmani et al., 2023; Peng et al., 2024; Wang et al., 2022b), and hybrid methods (Kumar et al., 2024a; Aldhyani et al., 2020). They can also be classified into temporal models (i.e., time series) (Shuvo et al., 2021) and spatio-temporal methods (Chang et al., 2016), which take into account local and/or global spatial features (Shao et al., 2020). Finally, they can address either short-term (Vlahogianni et al., 2014, 2004) or long-term times. Table 1 shows an overview of

these methods, and the most used acronyms in this paper are grouped in Table A.8 of Appendix A. For readers interested in the definitions of the full acronyms, please refer to the original references in the bibliography.

In statistics, linear time series models have been the first models employed for traffic forecasting. In 2003, Williams & Hoel (2003) proposed an Auto-Regressive and Integrated Moving Average (ARIMA) model, but it cannot achieve enough prediction accuracy for future traffic prediction. To address this limitation, Yang et al. (2021) proposes a network traffic prediction method based on simulated annealing optimized ARIMA-BPNN (Back Propagation Neural Network), making comprehensive use of the linear model ARIMA, the non-linear model BPNN, and the optimization algorithm Simulated Annealing. ARIMA-BPNN outperformed traditional linear and non-linear network traffic prediction models, with several improvements in network traffic prediction accuracy. Nevertheless, ARIMA-BPNN lacks to handle spatio-temporal forecasting problems. However, these autoregressive models are based on the stationary assumption of sequential data, which fails to hold in real traffic conditions that vary over time.

Conventional learning methods, including Support Vector Regression (SVR) (Wu et al., 2004) have also been used to address traffic flow prediction. Nevertheless, these shallow models employ hand-crafted features and do not fully explore complex spatio-temporal patterns, limiting their performance. Since the rise of deep learning, forecasting accuracy has been improved, as well as the management of spatio-temporal forecasting problems. For example, Jiang & Zhang (2019) modeled the whole city as a grid and applied a convolutional neural network (CNN). Transformer-based approaches have also been employed, but can be computationally costly and often fail to capture the detailed spatial correlations within traffic networks. Thus, both the CNN-based (Dai et al., 2017; Wu & Tan, 2016; Jiang & Zhang, 2019) and the transformer-based (Omar et al., 2024; Lu et al., 2024; Xiao & Chen, 2024; Lu, 2024; Zhao et al., 2024a; Pu et al., 2024; Liu & Zhang, 2024a) approaches are not optimal for traffic forecasting problems that have a graph-based form (Guo & Xie, 2021; Tang et al., 2021). This fact is clearly illustrated by the difference between WaveNet (Oord et al., 2016) and Graph-WaveNet (Wu et al., 2019). Transformer-based models provide good performance in

temporal dynamics but can be computationally costly and often fail to capture the detailed spatial correlations within the traffic network. To address traffic forecasting in a graph-based form, different graph-based methods have been used in the literature:

Table 1: Traffic Forecasting: A Representative Overview (Part 1).

Categories	**Sub-Categories**	**Methods** (Authors, Date)
Statistical Methods	Parametric Methods	History Average Model (HA) (Smith & Demetsky, 1997), Holt-Winters Model (Kochetkova et al., 2023)
Temporal Models		ARIMA (Williams & Hoel, 2003), STARIMA (Kamarianakis & Prastacos, 2005), ARIMA-BP (Yang et al., 2021)
(Shuvo et al., 2021) (Survey)		SARIMA-GCN-LSTM (Yang et al., 2024)
	Non-parametric Methods	KNN (Clark, 2003), ST-KNN (Cheng et al., 2018)
Machine Learning Models	Neural Networks	***MLP:*** STHMLP (Qin et al., 2023)
		RNN: CiRNN (Baruah & Organero, 2024)
		GRU: GRU (Shiang et al., 2020), BiGRU (Wang et al., 2022c) , ST-BiGRU (Zhang et al., 2024a)
	Support Vector Machines	LS-SVM (Zang & Liu, 2009), SVR (Wu et al., 2004), Linear SVR (Jin et al., 2007), TRU-VAR (Schimbinschi et al., 2017)
		Short-Term: VMD-SMA-SVR (Qu et al., 2024)
	Deep Neural Networks	***AE:*** SAE (Lv et al., 2015), DBN (Huang et al., 2014), TDANet (Li et al., 2023c)
	Temporal Models	***CNNs:*** CNN-LSTM (Dai et al., 2017), 1D-CNN (Wu & Tan, 2016)
	(Wang et al., 2022b) (Survey)	***Multi-Frequency Multi-Channel:*** ADMNet (Yuan et al., 2025)
	Spatio-Temporal Models	***LSTM:*** LSTM (Wang et al., 2021), LTSM-M (Tian et al., 2018), T-LSTM (Mou et al., 2019), MB-LSTM (Qu et al., 2022)
	(Lana et al., 2018) (Survey)	HierAttnLSTM (Zhang, 2024), DeepLSTM (Yu et al., 2017)
		Transformers: Encoder-Decoder Transformer (Omar et al., 2024), STPSformer (Lu et al., 2024)
		DSFormer-LRTC (Zhao et al., 2024a), MVSTT (Pu et al., 2024)
		DSTET (Sun et al., 2023), ADCT-Net (Kong et al., 2024)
		STHGFormer (Li et al., 2024b), TSANet (Kumar et al., 2024b), STGM (Lablack & Shen, 2023), ASTRformer (Wang et al., 2025)
	Large Language Models	STLLM-DF (Shao et al., 2024), R2T-LLM (Guo et al., 2024b)
	(Zhang et al., 2024g) (Survey)	LSGLLM-E (Rong et al., 2024), ST-LLM (Liu et al., 2024b)

– **Graph Convolution Neural Networks (GCNs):** The GCN-based methods obtain state-of-the-art performance, but their main limitation is that spatio-temporal GCN-based methods ignore the natural hierarchical structure of traffic systems, which is composed of the micro layers of road networks and the macro layers of region networks, in which the nodes are obtained through pooling methods and could include some hot traffic regions such as downtown. Most of the Spatio-Temporal Graph Convolutional Network-based methods (STGCN-based methods) ignore the relationship and role of the existing global and local road network graphs, except for the works (Guo et al., 2024a; Zhang et al., 2023b). Practically, the various improvements of GCN-based models can be classified into spatio-temporal models, dynamic models, and attention models, as can be seen in Table 2.

- **Graph Neural Networks (GNNs):** GCNs are the base model for all GNN-based approaches. Thus, GNNs are well-suited to traffic. For example, a road network is naturally a graph, with road intersections as the nodes and road connections as the edges. With graphs as the input, several GNN-based models have demonstrated superior performance to previous approaches on tasks, including road traffic flow and speed forecasting problems Garcia-Siguenza et al. (2023); Bai et al. (2020). But several recent methods have been published in 2024. However, Spatial-Temporal Graph Neural Networks (STGNNs) research focuses on aggregating short-term traffic history to predict future data.

- **Graph Transformer Networks (GTNs):** In 2024, graph transformer networks were introduced in traffic forecasting. For example, Chen et al. (2024c) designed a convolution-augmented graph sampling transformer network by using a hybrid graph transformer module based on the graph convolutions to perceive local structural details, which are incorporated into the transformer to augment the attention computation for more accurate global dependency modeling. In other work, Cai et al. (2024) developed a long-term correlations dual-graph transformer whilst Wei et al. (2024) used a transformer-based spatio-temporal graph diffusion convolution network. In a more general framework, Xiong et al. (2024) proposed a generalized spatial-temporal regression graph convolutional transformer. All the GTN models (LCDFormer (Cai et al., 2024), TSGDC (Wei et al., 2024), GSTRGCT(Xiong et al., 2024), CASTNet (Chen et al., 2024c)) outperform conventional GNN models.

- **Graph Large Language Models (GLLM):** In 2024, Moghadas et al. (2024b) proposed a graph-aware LLM for traffic prediction that considers proximal traffic information, taking spatial dependencies into account by encoding the network graph implicitly, especially without any prompting. The proposed method, named STRADA-LLM, outperforms STSGCN (Song et al., 2020) and GMAN (Zheng et al., 2020a)) published in 2020.

For more references, the reader can refer to the "Traffic Forecasting Website"[1], which groups more than 200 papers and codes in the field.

Table 2: Traffic Forecasting: A Representative Overview (Part 2).

Categories	**Methods** (Authors, Date)
Graph Convolutional Networks (56 papers)	***Spatio-Temporal GCN:*** STGCN (Yu et al., 2018), SLCNN (Zhang et al., 2020), ST-GCN (Agafonov, 2020), TSSRGCN (Chen et al., 2020a)
Classified by Mechanisms	LSTGCN (Gou et al., 2023), FDST-GCN (Zhang et al., 2023a), PaSTG (He et al., 2024)
	Attention Mechanism: ASTGCN (Guo et al., 2019), TSSRGCN (Chen et al., 2020a), A3T-GCNN (Bai et al., 2021), MGSTA (Li et al., 2024e)
	QA-STGACN (Qiu et al., 2024), TC-GCN (Wang et al., 2024e)
	Adaptive Mechanism: AdpSTGCN (Zhang et al., 2024d), ADSTGCN (Cui et al., 2023), AIMST (Zhang et al., 2024e)
	Bayesian: BSTGCN (Fu et al., 2020)
	Probabilistic: PSTGCN (Atkia & Naushin, 2024)
	Contrastive Learning: CDAGF (Ji et al., 2024), SCPT (Prabowo et al., 2024), STS-CCL (Li et al., 2024d), COGCN (Guo et al., 2024a)
	Correlation: STC-DGCN (Ge et al., 2024)
	Dynamic Mechanism: DGCNN (Diao et al., 2019), DST-GCNN (Chen et al., 2020b), STDE-DGCN (Zhang et al., 2022b),
	DDGCRN (Weng et al., 2023), DMGAN (Li et al., 2023b), DMGSTCN (Qin et al., 2024), FDST-GCN (Zhang et al., 2023a)
	iDCGCN (Liao et al., 2022), ADGCN (Isufi & Mazzola, 2021), STIDGCN (Liu & Zhang, 2024b)
	Frequency Domain: GDFDGCN (Li et al., 2024f)
	Generic: GDGCN (Xu et al., 2023)
	Geo-Distributed: GAP-LSTM (Altieri et al., 2024)
	Heterogeneous: HSTGCN (Zhong et al., 2021), HASTGCN (Mao et al., 2024)
	Hierarchical: HGCN (Guo et al., 2021a)
	Missing Data RT-GCN (Liu et al., 2024d), RPSTGCN (Akila & Nower, 2024)
	Multi-Graph: TFGAN (Khaled et al., 2022)
	Progressive Mechanism: PGCN (Shin & Yoon, 2024), PMGCN (Li et al., 2023d)
	Feature-Fusion: STFF-GCN (Shen & He, 2023)
	Hypergraph: MT-HGCN (Wang et al., 2022a), MDSTGCN (Liu et al., 2024c)
	Knowledge: HDMGCN (Qian et al., 2024), KaTaGCN (Wang et al., 2024f)
	Local Spatial: GSTPRN (Zhang et al., 2023b)
	Local/Global Spatial: LGSTGN (Zong et al., 2024), GASTN (He et al., 2022), ST-GDN (Zhang et al., 2021),
	ASTGCNs (Chen et al., 2023), **LSTGAN (Holistic) (Ours)**
	Recurrent:AGCRN (Bai et al., 2020), DAGCRN (Shi et al., 2023), TAGRN (Shao et al., 2023)
	Synchronous: STSGCN (Song et al., 2020), ABSGCN (Shen et al., 2021)
	Long-Term: MA-GCN (Peng & Zhang, 2023), GTADGCN (Wang et al., 2024a), **LSTGAN (Holistic) (Ours)**
	Short-term/Long-Term: Graph CNN-LSTM (Bogaerts et al., 2020)
Graph Convolutional Networks/GRU (2 papers)	***Hybrid Models:*** STBGRN (Zhang et al., 2024a), GCST-GRU (Hussain et al., 2023)

2.2. *Graph-based Models: Their Limitations*

Even if GNN-based models demonstrate high prediction performance and can be employed in federated (Pavlidis et al., 2024; Zhang et al., 2022a; Hu et al., 2024)

[1] `https://sites.google.com/view/trafficforecasting/home`

Table 3: Traffic Forecasting: A Representative Overview (Part 3).

Categories	**Methods (Authors, Date)**
Graph Neural Networks (55 papers)	***Spatio-Temporal GNN:*** Gated GNNs (Chen et al., 2019), STGNN (Wang et al., 2020b)
Classified by Mechanisms	STMAG (Wang et al., 2020a), ST-MGAT (Tian et al., 2020), GWGR (Cui et al., 2020a)
(Jiang & Luo, 2022) (Survey)	GMN (Cui et al., 2020b), CGProNet (Einizade et al., 2024b), CITRUS (Einizade et al., 2024a)
(Rahmani et al., 2023) (Survey)	***Adaptive Mechanism:*** AJSTGL (Wang & Chen, 2024)
(Peng et al., 2024) (Survey)	***Attention Mechanism:*** ConstGat (Fang et al., 2020), GR-FAST (Li et al., 2024a), TFM-GCAM (Chen et al., 2024a)
	HMIAN (Sun et al., 2022), DFAGCN (Liu et al., 2024e), STBGAT (Lakmal et al., 2024)
	Contrastive Learning: USTGCL (Pan & Ren, 2024)
	Conjoint Perspective: COOL (Ju et al., 2024)
	Differential Equations: HSTGODE (Xu et al., 2025), STDDE (Long et al., 2024), GPDE (Bao et al., 2024)
	NCDE (Wang & Zang, 2023), STG-NRDE (Choi & Park, 2024), ST-DGDE (Zhou et al., 2023), STHODE (Yao et al., 2023)
	GAPDE (Almousa & Lee, 2024), GRAM-ODE (Liu et al., 2023b), CEGNCDE (Wu & Chen, 2024)
	DM-AGODE (Bai et al., 2023), FedGODE (Huthaifi et al., 2024), D^3STN (Zhu et al., 2024)
	Dynamic Mechanism: Dynamic-GRCNN (Peng et al., 2020), ASTGN (Guo et al., 2021b), DTPO-GNN (Waikhom et al., 2023)
	Heterogeneous: FEHGCARN (Zhang et al., 2024c)
	Hierarchical: HSeq2Seq (Xie et al., 2024)
	Frequency Domain: D2STGNN (Liu et al., 2024a)
	Periodic/Aperiodic TPDGCN (Zhang et al., 2024b), DualCast (Su et al., 2024)
	Pretrained Local Augmentation: LAM-STGNN (Yao et al., 2024)
	Residual: STRGAT (Zhang et al., 2023c)
	Uncertainty Quantification: CreST (Zhou et al., 2024), STUP (Jin et al., 2024)
	Variational: VGRAN (Zhou et al., 2020)
	Low-complexity Algorithms: BigST (Han et al., 2024), ST-MLP (Wang et al., 2023), VQGG (Ke et al., 2024)
	Missing Data: Di-GraphGAN (Li et al., 2024c), TPB (Liu et al., 2023c), MTPB (Liu et al., 2024g), FEPCross (Liu et al., 2024f)
	Transfer Learning: MegaCRN (Jiang et al., 2023b), TransGTR (Jiang et al., 2023a), STF (Kawabata & Toda, 2024)
	Multi-Graph/Hypergraph: MGHCN (Fan et al., 2025), PAHNN (Zhao et al., 2024c), DHSL (Wang et al., 2024d)
Graph Transformers Networks (4 papers)	LCDFormer (Cai et al., 2024), TSGDC (Wei et al., 2024)
	GSTRGCT (Xiong et al., 2024), CASTNet (Chen et al., 2024c)
Graph Large Language Models (1 paper)	STRADA-LLM (Moghadas et al., 2024b)

and distributed (Mostafi et al., 2024; Nazzal et al., 2024) manners, they present the following limitations:

1. Error accumulation and the vanishing gradient problem make temporal neural networks less effective at handling long time series. Consequently, GNN-based models often struggle to produce reliable results for multi-step predictions.
2. The ability of GNNs to manage spatial correlations is dependent on road topological connections, making them unable to identify geo-parcel-based traffic patterns that are independent of road networks.

3. GNNs can only capture spatio-temporal dependencies between pairs of nodes, which cannot handle high-order correlations among time series. To address this limitation, multi-graph (MGHCN (Fan et al., 2025)) and hypergraph approaches (PAHNN (Zhao et al., 2024c), DHSL (Wang et al., 2024d)) have been proposed.
4. The message passing in GNNs is immediate, whilst the spatial message interactions among neighboring nodes are delayed in reality. Indeed, the change of traffic flow at one node will take several minutes (i.e., time delay) to influence its connected neighbors. To address this issue, differential equation models have been used in the literature (Xu et al., 2025; Long et al., 2024; Bao et al., 2024; Wang & Zang, 2023; Choi & Park, 2024; Zhou et al., 2023). It includes both delay effects and continuity into a unified delay differential equation framework. In 2025, Zheng et al. (2025) propose a Delay-aware Directed Graph Attention (DDGA) to delay information propagation between nodes.
5. Traffic conditions undergo continuous changes. The prediction frequency for traffic flow forecasting may vary based on specific scenario requirements. Thus, adaptive (AJSTGL (Wang & Chen, 2024)), dynamic (Dynamic-GRCNN (Peng et al., 2020), ASTGN (Guo et al., 2021b), DTPO-GNN (Waikhom et al., 2023)), and time-varying (HTVGNN (Dai et al., 2024)) mechanisms are required.
6. STGNNs neglect the incorporation of the cyclical/periodical patterns that appear in the traffic historical data. For example, the cyclical/periodical patterns of traffic on the same day or hour of each week or season of each year can help improve the accuracy of future traffic predictions. To leverage periodicity information with recognizing that periodic patterns come from prior ones-a phenomenon, Zhang et al. (2024b) defined the transitivity of periodicity and designed a Dynamic Graph Convolutional Network grounded in Transmissibility-Periodicity (TPDGCN).
7. STGNNs provide forecasts without estimates of data and model uncertainty, which are critical for understanding inherent variations of the data and forecast limitations due to a lack of training data. In 2024, Mallick et al. (2024) developed a scalable deep ensemble approach to quantify data and model uncertainties for STGNNs.

8. The generalization of STGNNs in extended temporal scenarios and cross-city applications remains largely unexplored. In 2024, Wang et al. (2024c) evaluated several models and demonstrated performance degradation in existing STGNNs over time due to their limited inductive capabilities. To address this limitation, Wang et al. (2024c) proposed a Principal Component Analysis (PCA) embedding approach that allows the models to adapt to new scenarios without retraining.
9. Graph indistinguishability requires investigations, as graphs learned by existing methods tend to converge to implicit and indistinguishable representations, deviating from the genuine distribution. This problem is due to the lack of three factors within graphs: the intrinsic graph features, the temporal-distinct features, and the node-distinct features. In 2024, Chen et al. (2024b) developed a Temporal-Aware Structure-Semantic Coupled Graph Network (TASSGN) that allows for tackling these three factors.
10. The prior STGNNs ignore the natural hierarchical structure of the traffic road networks and thus fail to capture the macro-spatial dependence of region networks. To address this limitation, Xie et al. (2024) developed a hierarchical GNN for accurate mobile traffic forecasting that fully exploits the cross-regional feature impacts.
11. STGNNs require prohibitive quadratic computational complexity to capture long-range spatio-temporal dependencies, thus limiting their use to long historical sequences on large-scale road networks in the real world (Han et al., 2024).
12. The presence of noise in graph nodes can significantly degrade the performance of GNNs (Ding et al., 2024), particularly when local spatial or spectral features are unreliable. To address this issue, several approaches have been proposed. For instance, MRGAT (Ding et al., 2023) introduces multi-scale receptive fields and edge-aware attention mechanisms to enhance local-global feature extraction in noisy environments. AF2GNN (Ding et al., 2022) improves robustness by adaptively combining multiple graph filters and aggregators. MARP (Zhang et al., 2023d) employs an adaptive path aggregation mechanism to dynamically learn receptive fields and mitigate the impact of noisy nodes on classification.

In addition, STGCNs and STGNNs models significantly decrease their performance in addressing traffic out-of-distribution (OOD) scenarios. In 2024, Wang et al. (2024b) demonstrates that this loss is due to the models' inability to adapt to previously unobserved spatial relationships. Thus, different strategies have been employed in literature to address these limitations, and they mainly addressed spatio-temporal models, dynamic models, and attention models, as can be seen in Table 3. To address issues 1) and 2), several studies incorporate dynamic graphs to convey additional road network information, but challenges, including limited flexibility and adaptability, continue to exist. Other works employ the attention mechanism, allowing for a dynamic assessment of the correlations between target sequences and selectively concentrating on pertinent content. For example, to address temporal dependencies, Liao et al. (2022) leveraged the proficiency of the attention mechanism in managing long sequences, thus improving the accuracy of long-term traffic forecasting. Regarding spatial dependencies, Zheng et al. (2020a) introduced a multiheaded spatial attention mechanism to discern various spatial correlations. However, the attention mechanism is predominantly employed as a module within forecasting models rather than being utilized independently. For example, models such as Attention based Spatial-Temporal Graph Convolutional Networks (ASTGCN) (Guo et al., 2019) integrate GNNs with the attention mechanism takes into account spatial dependencies. But the attention mechanism, designed for sequential data, is ill-suited to address road network graphs featuring non-Euclidean data types. Thus, the attention mechanism must be employed with GNNs to effectively capture spatial dependencies. However, these methods allow the forecasting models to synthesize spatial understanding by integrating the attention mechanism with GNNs. In essence, the models that independently assess dependencies arising from traffic patterns and road networks overlook their interrelation and thus fail to construct a cohesive urban spatial perspective. Furthermore, the employment of two distinct models for spatial correlation mining introduces additional computational times and requires unnecessary costs. To address both local and global spatial influences in GNNs, He et al. (2022) proposed to predict mobile traffic demand. This method uses a model that extracts local data within a region and represents the extracted information as nodes. Then, another model extracts global information from

these nodes. In another work, Zhang et al. (2021) considered the regions as nodes in the graph. They apply an attention mechanism to the graph, followed by a graph convolution operation (GCN) sequentially, and make predictions by a simple prediction layer. Chen et al. (2023) took into account the local information as an aggregation of each time step, while global information is the aggregation of data across the entire timeline.

2.3. Our Contributions

Despite the numerous works, no one algorithm is able to simultaneously address all the key challenges that are present in traffic forecasting during long sequences requiring specific mechanisms (Zhang et al., 2024f). In this context, we decide to focus on a long-term spatio-temporal graph attention network for traffic forecasting named LSTGAN. Its robustness in scenarios with incomplete data and also its generalization on diverse urban layouts (e.g., sparse sensor networks) will be investigated in future work, as both of these points require extensive work leading to another article. Inspired by research on attention-based graph representation, our work introduces a more holistic understanding of temporal (i.e., long-term) and spatial (i.e., local and global) influences to tackle the limitations inherent in the conventional attention mechanism in long-term scenarios.

Our model is different from previous works addressing local and global issues (LGSTGN (Zong et al., 2024), GASTN (He et al., 2022), ST-GDN (Zhang et al., 2021), ASTGCNs (Chen et al., 2023)) in that we consider local information as the aggregation of data from neighboring nodes, while global information is the aggregation of all nodes in the graph, each according to its attention weight, even if they are not directly adjacent. Additionally, we consider nodes as road traffic sensors, not regions. The extracted local and global data, along with historical data, are concatenated side by side (in parallel, not sequentially), and an attention-based decoder then analyzes this data to identify which pieces will have the most significant impact on the forecasting.

3. Proposed Model

In this section, we provide a formal statement of our problem and present the network architecture of our model. Then, we detail our proposed spatio-temporal unit and the forecasting unit.

3.1. Problem Definition

We consider a network of roads equipped with a set of sensors that capture the speed of vehicles. Traffic forecasting is the estimation of future traffic flow based on past traffic flow data collected from the different sensors that are deployed on the road network. The topological form of the distribution of road traffic sensors can be represented by a graph $\mathbf{G}(\mathbf{V}, \mathbf{E}, \mathbf{A})$. We denote by $\mathbf{V}$ a set of N nodes, each node represents a sensor of the road network, and by $\mathbf{E}$ the set of edges between the nodes represented by the spatial adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$. We represent the traffic signals on the nodes at time t as $\mathbf{x^t} \in \mathbb{R}^{N \times C}$, where C is the feature dimension. Given the historically observed P time steps of traffic signals, denoted by $\mathbf{X} = \{x^{t_1}, x^{t_2}.., x^{t_p}\} \in \mathbb{R}^{P \times N \times C}$, our goal is to predict the next H time steps of traffic signals $\mathbf{Y} = \{x^{t_{p+1}}, x^{t_{p+2}}.., x^{t_{p+h}}\} \in \mathbb{R}^{H \times N \times C}$.

3.2. Network Architecture

Our model consists of two main parts. The first part is the spatio-temporal unit, designed to extract spatial and temporal features of sensors from long-term historical data. This unit consists of four modules: two modules for processing weekly and daily temporal data, and two modules for processing local and global spatial data. The idea of using temporal modules is based on the hypothesis that usually, people's traffic habits can be observed on a daily (e.g., rush hours of a day) and weekly basis (weekend). Regarding the spatial modules, sensors are affected by signals from nearby sensors.

These modules then extract spatial and temporal patterns, integrate them to create a spatio-temporal representation denoted as $\boldsymbol{STe} \in \mathbb{R}^{P \times N \times d}$, where d represents the dimension of the integrated features from the spatio-temporal modules, and then pass this representation to the forecasting unit, which uses a self-attention decoder to generate accurate outputs for future traffic forecasts. The general architecture of our model is illustrated in Figure 1.

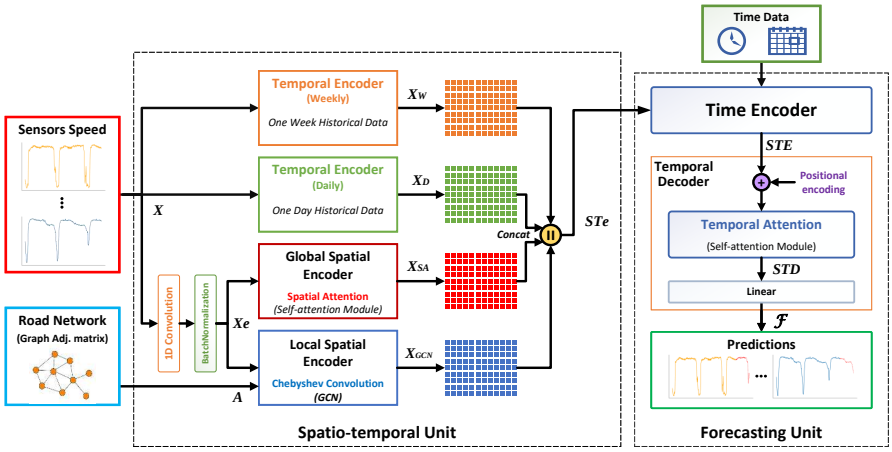


Figure 1: Model Architecture.

3.3. Spatio-Temporal Unit

This unit processes long-term temporal historical data through a weekly module to handle data from one week before the prediction time, a daily module to process data from one day before the prediction, and two modules to process spatial dependencies for the last hour preceding the prediction time: one employing graph convolution and the other using self-attention mechanisms. The output of these modules is fused into a single feature map that can be used for forecasting purposes. Figure 2 provides an illustrative example of how local and global spatial dependencies are aggregated for the target sensor (red node). Local dependencies (green dashed lines) are aggregated using a Graph Convolutional Network (GCN), capturing information from immediate neighboring sensors in the green zone, while global dependencies (purple dashed lines) are modeled using a self-attention mechanism, allowing the sensor to selectively integrate information from distant sensors across the road network. The line thickness (thick, normal, or thin) represents the attention score assigned to each node, indicating its relative importance in the aggregation process.

3.3.1. Long-Term Temporal Dependencies

To deal with long-term historical data, we created a temporal encoder module to encode this data by decreasing the size of the inputs and increasing the number of

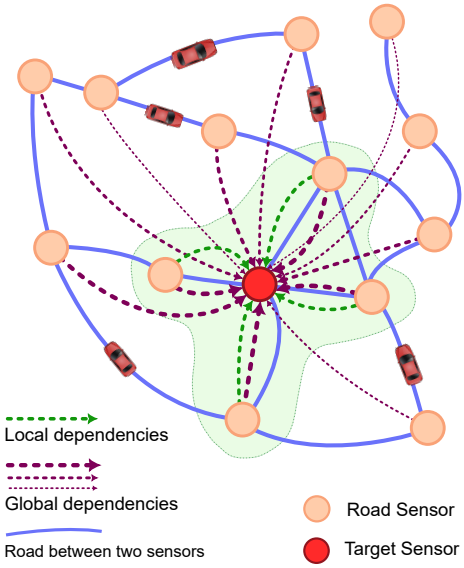


Figure 2: Aggregation of local and global spatial dependencies.

feature channels, a process referred to as *downsampling*, the same principle used in the U-Net architecture (Ronneberger et al., 2015).

Traffic data exhibit strong spatial and temporal correlations. It also contains periodic patterns such as daily and weekly trends, as well as local fluctuations such as rush hours. To effectively capture these dependencies, we used convolutional downsampling for its ability to reduce dimensionality while preserving important information and filtering out noise and redundant variations in traffic patterns. Furthermore, it provides a hierarchical representation by gradually reducing resolution, which allows the model to capture high-level spatial and temporal dependencies while retaining fine-grained details. In addition, the downsampling enhances computational efficiency over direct sequence modeling, reducing complexity and making real-time forecasting more feasible.

The downsampling process is performed twice to achieve the desired output size, where each occurrence consists of two 1D convolutional layers. The first convolutional layer has a kernel size of 3 and padding to maintain the same size as the input, it operates on an input tensor $\mathbf{S} \in \mathbb{R}^{T \times N \times C_1}$, where T represents the temporal dimension

(number of time steps), N is the number of nodes, and C_1 is the number of input feature channels. This layer produces an output tensor $\mathbf{S}' \in \mathbb{R}^{T \times N \times C_2}$, where C_2 represents the transformed feature dimension after applying the convolution. This layer is followed by the ReLU activation function.

The second convolutional layer has a kernel size of k that determines the receptive field, ensuring that a sufficient number of past time steps contribute to each output value, and a stride of s that controls how the convolutional filter moves across the input sequence to effectively reduce the input resolution. The second convolution layer operates on the output of the previous layer, taking as input the tensor $\mathbf{S}' \in \mathbb{R}^{T \times N \times C_2}$, and produces an output tensor $\mathbf{S}'' \in \mathbb{R}^{T' \times N \times C_3}$, where T' represents the reduced temporal dimension calculated by Eq. 1 and C_3 is the updated number of feature channels. This convolution operation is represented by Eq. 2.

$$T' = \left\lfloor \frac{T - k}{s} \right\rfloor + 1 \tag{1}$$

$$\mathbf{S}''[n] = \sum_{m=0}^{k-1} \mathbf{S}'[n \cdot s + m] \cdot \mathbf{W}[m] \tag{2}$$

Where $\mathbf{S}''[n]$ is the n-th output value after applying the convolution, $\mathbf{W}[m]$ is the m-th element of the convolution kernel $\mathbf{W}$, and $\mathbf{S}'[n \cdot s + m]$ is the input vector accessed at the position determined by the current stride and kernel index. k is the size of the convolution kernel $\mathbf{W}$, and $\lfloor \cdot \rfloor$ represents the floor function.

We used a kernel size k equal to stride s to ensure that feature extraction does not overlap and prevents redundancy while preserving key information. In addition, this choice improves computational efficiency by reducing redundant computations and consequently reducing computational complexity.

The stride s changes according to each downsampling block and also whether the inputs are for the weekly or daily data. After each downsampling block, we apply batch normalization, with ReLU applied between the two blocks. Note that each convolutional layer increases the number of feature channels. The temporal encoder produces an output tensor $\mathbf{X_D} \in \mathbb{R}^{P \times N \times C_d}$ for daily data and an output tensor $\mathbf{X_W} \in \mathbb{R}^{P \times N \times C_w}$ for weekly data, where P represents the historically observed time steps of traffic speeds,

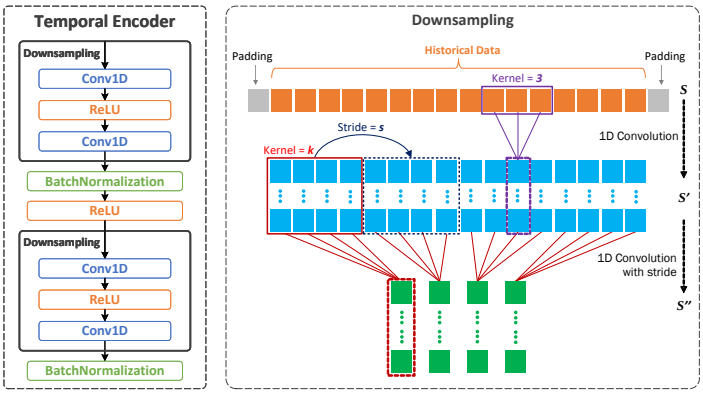


Figure 3: Architecture of the temporal encoding module.

N denotes the number of sensors, and C_d and C_w are the dimensions of the resulting tensors for daily and weekly data, respectively. Figure 3 shows the architecture of the temporal encoder module and provides an illustration of the downsampling process.

3.3.2. Spatial Dependencies

In real-world traffic scenarios, the traveling speed on a particular road is spatially affected by the traveling speeds on neighboring roads. To capture these spatial dependencies, we employ two distinct modules. The first module, the global spatial encoder, utilizes a self-attention mechanism to capture global node-to-node dependencies across the graph. This mechanism captures interactions between distant or non-adjacent nodes, allowing for a broader understanding of the spatial dynamics. The second module, the local spatial encoder, employs graph convolution to learn node features by aggregating information from local neighbors based on the graph's structure. By combining these modules, our approach ensures that both local and global information is captured, prioritizing the most relevant features. We feed these modules with the data collected over the last hour, as this recent information directly impacts the traffic conditions that we aim to predict for the upcoming hour.

To better extract the spatial hidden traffic features, we use a convolutional layer with a kernel size equal to 1 followed by batch normalization to obtain the traffic feature embedding. The operation is represented by Eq. 3. The input to this layer is the sensor

signals denoted by $\mathbf{X} \in \mathbb{R}^{P \times N \times C}$ where P represents the historically observed time steps of traffic speeds, and C is the feature dimension (in our case, $C = 1$, as the only feature is traffic speed). The output is $\mathbf{Xe} \in \mathbb{R}^{P \times N \times C_s}$, where C_s is the embedding dimension.

$$\mathbf{Xe} = BatchNorm\Big(Conv\big(\mathbf{X}^{1:P}\big)\Big) \tag{3}$$

We then pass this output embedding to the global spatial encoder and the local spatial encoder to extract valuable spatial dependencies.

Global Spatial Encoder. To capture global spatial dependencies, we use the self-attention mechanism (Vaswani, 2017), which enables each node to dynamically pay attention to every other node in the graph, regardless of their spatial proximity. The self-attention mechanism captures interactions between nodes by enabling each node to learn from all other nodes in the graph.

For each node, three distinct vectors, query ($\mathbf{q}_i$), key ($\mathbf{k}_i$), and value ($\mathbf{v}_i$) are generated through a linear transformation that utilizes the learned weight matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$. These vectors are linear transformations of the original feature vector $\mathbf{Xe}_i \in \mathbb{R}^{P \times C_s}$ of node i [2], and are computed as shown in Eq. 4.

$$\mathbf{q}_i = \mathbf{W}_Q \mathbf{Xe}_i, \quad \mathbf{k}_i = \mathbf{W}_K \mathbf{Xe}_i, \quad \mathbf{v}_i = \mathbf{W}_V \mathbf{Xe}_i \tag{4}$$

The attention score $\boldsymbol{\alpha}_{ij}$ between node i and node j is computed using the scaled dot-product between the query of node i and the key of node j, and then normalized using the softmax function to ensure that the sum of the attention scores for a given node across all other nodes equals 1, as shown in Eq. 5.

$$\boldsymbol{\alpha}_{ij} = \text{softmax}\left(\frac{\mathbf{q}_i \cdot \mathbf{k}_j^T}{\sqrt{d_k}}\right) \tag{5}$$

Where d_k is the dimensionality of the key vectors. The resulting normalized attention scores $\boldsymbol{\alpha}_{ij}$ serve as weights that indicate how much attention node i should pay to node

[2]The dimension P, which represents the number of time steps considered for encoding temporal dependencies, is preserved. Each time step is processed along with its corresponding time steps in other node tensors.

j. Using these attention weights, a weighted sum of the value vectors $\mathbf{v}_j$ of all N nodes is computed, producing the updated feature representation $\mathbf{Xe}'_i \in \mathbb{R}^{P \times C_s}$ for node i, as shown in Eq. 6.

$$\mathbf{Xe}'_i = \sum_{j=1}^{N} \boldsymbol{\alpha}_{ij} \cdot \mathbf{v}_j \tag{6}$$

After the scaled dot-product is computed, the resulting value is linearly transformed using the learned weight matrix $\mathbf{W_o}$ to produce the final self-attention output $\mathbf{Xe^*_i} \in \mathbb{R}^{P \times C_s}$. This is followed by a feed-forward network comprising two linear transformations, with a ReLU activation function applied in between. Residual connections are incorporated around both the self-attention mechanism and the feed-forward network, followed by layer normalization to stabilize the training process. Figure 4 shows the architecture of the self-attention module.

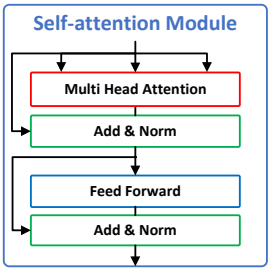


Figure 4: Architecture of the self-attention module.

We extend the attention mechanism to multi-head attention to stabilize the learning process, allowing the model to focus independently on different spatial relationships between the graph nodes and enabling parallel computation, thereby improving the training speed. In multi-head attention, the initial embedding dimension is decomposed into h heads. Each head is processed using the scaled dot-product attention. The outputs of all heads are then concatenated and projected through a linear transformation with the learned weight matrix $\mathbf{W_o}$ to produce the final output.

The result of the global spatial encoder is a refined feature vector of each node $\mathbf{X_{SA}} \in \mathbb{R}^{P \times N \times C_s}$. This is achieved by updating each node's feature to include relevant information from other nodes in the graph. This way, each node's final representation

incorporates the global context of the entire graph.

Local Spatial Encoder. To capture local spatial dependencies, we employed a Graph Convolutional Network (GCN), which provides an embedding for each node by aggregating the information of neighboring nodes in topological graph structures. In particular, we use the ChebConv layer (Defferrard et al., 2016), which generalizes convolutional neural networks (CNNs) to graph domains using spectral graph theory, retaining the efficiency and complexity of traditional CNNs while being universally applicable to any graph structure.

ChebConv is used to perform graph convolutions in the spectral domain by normalizing and then approximating the graph Laplacian using Chebyshev polynomials, which allows the model to capture information from a node's wider neighborhood. The output of the convolution operation, $\mathbf{X_{GCN}} \in \mathbb{R}^{P \times N \times C_l}$, where C_l represents the updated number of feature channels, is computed from the input $\mathbf{Xe} \in \mathbb{R}^{P \times N \times C_s}$ using Eq. 7.

$$\mathbf{X_{GCN}} = \sum_{k=1}^{K} Z^{(k)} \cdot \Theta^{(k)} \tag{7}$$

Where K is the Chebyshev filter size, which determines the order of the polynomial used to approximate the spectral graph convolution filter. Rather than performing exact spectral convolutions, which require eigen-decomposition of the graph Laplacian, ChebConv uses a truncated expansion of Chebyshev polynomials to efficiently approximate the filter. This approach enables localized filtering on the graph up to K-hop neighborhoods, allowing the model to capture spatial dependencies while avoiding the computational overhead of spectral operations. The filter size K directly influences the model's receptive field. Lower values focus on immediate neighbors, capturing fine-grained local patterns, while higher values incorporate broader spatial context by aggregating information from more distant nodes. Θ is a learnable weight. $Z^{(k)}$ is computed recursively by Eq. 8.

$$
\begin{aligned}
Z^{(1)} &= Xe \\
Z^{(2)} &= \hat{L} \cdot Xe \\
Z^{(k)} &= 2 \cdot \hat{L} \cdot Z^{(k-1)} - Z^{(k-2)}
\end{aligned} \tag{8}
$$

X is the input signal, $\hat{L}$ denotes the scaled and normalized Laplacian, computed by Eq. 9.

$$
\hat{L} = \frac{2L}{\lambda_{max}} - I \tag{9}
$$

where λ_{max} is the largest eigenvalue of L and I is the identity matrix.

3.4. Forecasting Unit

Traffic conditions on a road change nonlinearly based on observations of the previous condition of that road, as well as neighboring roads that may directly affect traffic on the same road. The data processed by the Spatio-temporal unit provides long-term historical data that captures traffic trends, as well as spatial data that includes information about neighboring roads. This data provides rich features that allow the model to understand temporal patterns and spatial relationships to make accurate predictions.

The objective of the forecasting unit is to generate traffic speed predictions for the upcoming hour by leveraging long-term historical data and spatial data processed by the preceding spatio-temporal unit. For this purpose, we utilized a self-attention-based decoder, a neural network module designed to address sequence-to-sequence problems.

Before using the decoder, we incorporate a Time Encoder module to enrich the input with explicit temporal context. This design is motivated by the inherent periodicity in traffic patterns, which tend to repeat daily and weekly due to regular human activities. As illustrated in Figure 5, traffic speed trends on Tuesdays and Saturdays show consistent patterns across consecutive weeks, both in timing and magnitude. This demonstrates the existence of periodic patterns in traffic behavior based on time.

3.4.1. Time Encoding

The resulting data $\boldsymbol{STe} \in \mathbb{R}^{P \times N \times d}$ from the previous Spatio-temporal unit contains only traffic speed data embeddings. To add time information, we used the Time Encoder module, illustrated in Figure 6. This module encodes the time information of

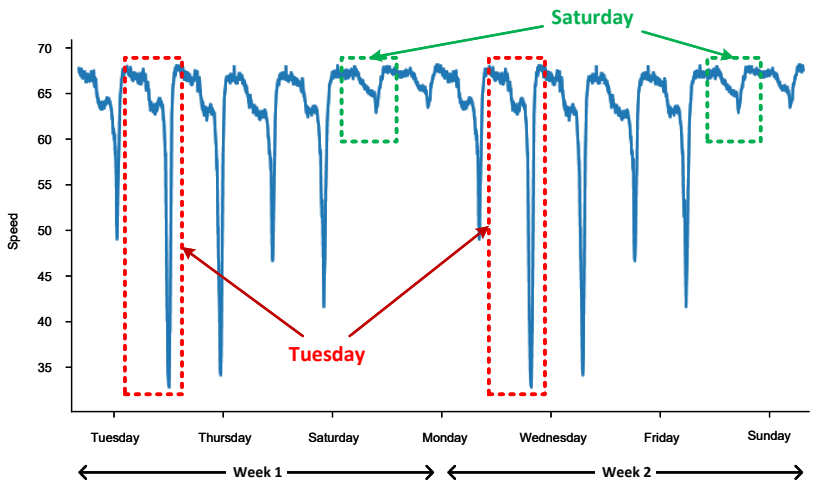


Figure 5: Traffic speed data collected over two consecutive weeks in the PEMS-Bay dataset.

each time step into a vector using a one-hot encoding scheme, chosen for its ability to represent discrete time categories without introducing any artificial ordering. This approach provides a simple, effective, and interpretable representation of temporal information.

The available time information includes the day of the week and the time of day. The day of the week is encoded in a vector $\boldsymbol{V^d}$ of length 7, representing the number of days of the week. The time of day is encoded in a vector $\boldsymbol{V^t}$ of the same length as the number of time steps in a day, which in our case is 288 [3].

The day of week vector $\boldsymbol{V^d}$ and time of day vector $\boldsymbol{V^t}$ are encoded using one-hot coding. For the day of week: let $\boldsymbol{V_{dow}}$ be a reference vector containing the seven days of the week, thus of size 7, and let *Day-of-week* be a day of the week to encode, then $\boldsymbol{V^d}[i] = 1$ if $\boldsymbol{V_{dow}}[i] = \textit{Day-of-week}$, 0 else.

For time of day: let $\boldsymbol{V_{tod}}$ be a reference vector containing the 288 times of a day and let *Time-of-day* be a time of day to encode (from 1 to 288, representing 5-minute

[3]Most traffic datasets are collected every 5 minutes, and a day contains 1440 minutes (24 hours $\times$ 60 minutes), Dividing 1440 by 5 minutes (the data collection rate) yields 288 time steps per day.

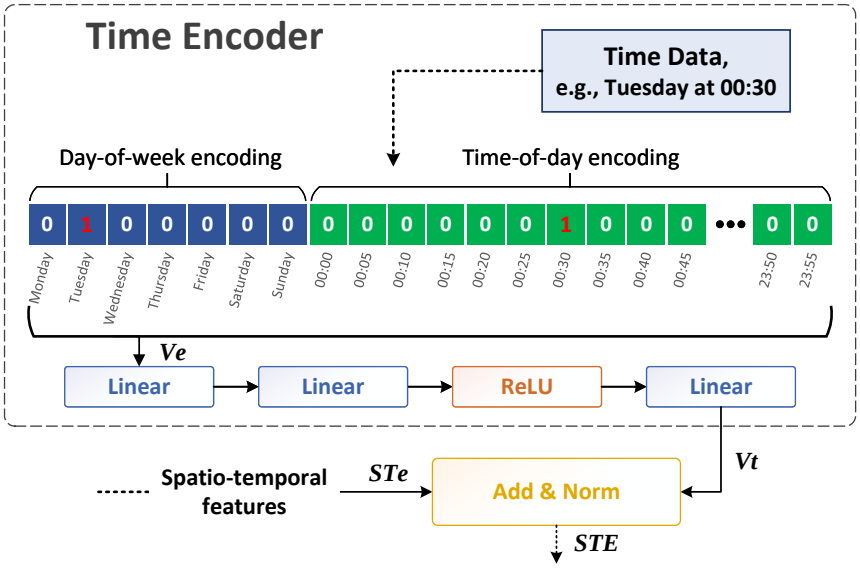


Figure 6: Time encoder module, with an example of encoding Tuesday at 00:30.

intervals), then $\boldsymbol{V^t}[i] = 1$ if $\boldsymbol{V_{tod}}[i] = \textit{Time-of-day}$, 0 else.

The two vectors are then combined to create a vector $\boldsymbol{Ve} = \boldsymbol{V^d} \oplus \boldsymbol{V^t}$ of length 295 [4].

The resulting vector $\boldsymbol{Ve}$ is passed through three *linear* layers, with *ReLU* activation functions in-between, to produce a time encoding output $\boldsymbol{Vt}$ of a length equal to the number of $\boldsymbol{STe}$ feature channels. Finally, the time encoding vector $\boldsymbol{Vt}$ and spatio-temporal data $\boldsymbol{STe}$ are fused and normalized to obtain integrated spatio-temporal data $\boldsymbol{STE} \in \mathbb{R}^{P \times N \times d}$ that includes time information, as represented by Eq. 10.

$$\boldsymbol{STE} = Norm\big(\boldsymbol{STe} + \boldsymbol{Vt}\big) \tag{10}$$

3.4.2. Temporal Decoder

The temporal decoder employs the same attention mechanism as the spatial encoder, with the difference that attention scores are computed across time steps for each

[4] 7 (days) + 288 (time steps) = 295.

node (sensor) instead of across spatial locations. This allows the model to weigh the importance of different time steps within the input sequence, thereby capturing long-range dependencies and temporal patterns in traffic speed data.

Given an input sequence $\mathbf{STE} = \{ste^1, ste^2, ..., ste^P\} \in \mathbb{R}^{P \times N \times d}$, which represents the historical data extracted from the Spatio-Temporal Unit and which includes time information, the decoder processes the entire sequence to generate the H future time step encodings $\mathbf{STD} = \{std^1, std^2, ..., std^H\} \in \mathbb{R}^{H \times N \times d}$.

First, positional encodings are applied to the input (Vaswani, 2017). This allows the model to understand the order of time steps by providing each time step with a unique representation that reflects its position within the sequence. The positional encodings are constructed using sinusoidal functions and then summed with input embeddings. The positional encoding for a time step t is defined by equations 11.

$$\mathrm{PE}_{(t,2i)} = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad \mathrm{PE}_{(t,2i+1)} = \cos\left(\frac{t}{10000^{2i/d}}\right) \tag{11}$$

Where i is the dimension of the positional encoding, and d is the dimension of the input embeddings.

Once the positional information is added to the sequence, the attention mechanism is applied by deriving the query, key, and value matrices from the input sequence and applying linear transformations to these matrices. Next, attention scores between time steps are computed using scaled dot-product attention, which determines the influence that each past time step should have on predicting future time steps. A feed-forward network is then used, and residual connections are added to improve the flow of information and ensure stable gradient updates during training.

Finally, the forecasting units conclude with a linear layer that transforms the decoded representations $\mathbf{STD} \in \mathbb{R}^{H \times N \times d}$ from the temporal decoder to generate the final prediction results $\mathcal{F} \in \mathbb{R}^{H \times N \times 1}$.

4. Experiments

4.1. Baselines

To determine the effectiveness of the proposed model, we compared its performance with several methods and models in the field of traffic forecasting, including

baseline prediction methods, models based on deep learning, and graph-based models that attempt to address also the spatio-temporal and long-term aspects. The baseline models include:

- **ARIMA** (Williams & Hoel, 2003) (Auto-Regressive Integrated Moving Average), which uses a Kalman filter to predict future values.
- **SVR** (Wu et al., 2004) (Support Vector Regression) employs a linear support vector machine for regression tasks.
- **WaveNet** (Oord et al., 2016) is a deep neural network architecture that uses sequential data.
- **FC-LSTM** (Karimzadeh et al., 2021) is a recurrent neural network with a fully connected LSTM hidden unit. These models do not require a predefined graph and rely only on sequential data.

The deep neural network models that utilize the spatio-temporal data extracted from the graphs that we compared our work with are the following:

- **STGCN** (Yu et al., 2018) uses graph convolution layers with a gated temporal convolution operation to capture spatio-temporal correlations.
- **DCRNN** (Li et al., 2018) combines diffusion convolution on graph and recurrent units (GRU) to make the spatial-temporal prediction.
- **ASTGCN** (Guo et al., 2019) uses attention mechanisms and CNNs to capture spatio-temporal correlations.
- **Graph-WaveNet** (Wu et al., 2019) dynamically learns spatial and temporal dependencies using an adaptive dependency matrix and dilated convolutions, enabling effective spatial-temporal modeling.
- **GGRNN** (Ruiz et al., 2020) integrates the recurrent hidden states of RNNs with Graph Signal Processing (GSP) by merging the RNN architecture with graph convolutional layers.

- **DST-GCNN** (Chen et al., 2020b) employs a graph-based convolutional layer to extract spatio-temporal information from traffic flow data and integrates the learned graph structures to predict future flows.
- **GMAN** (Zheng et al., 2020a) proposes a spatio-temporal attention-based encoder-decoder architecture to predict future traffic conditions by modeling the complex relationships between time and location factors.
- **TransGAT** (Wang et al., 2022d) aims to enhance traffic flow forecasting by combining dynamic attention for spatial features with a gated temporal convolutional network.
- **SDGCN** (Li et al., 2023a) employs attention fusion networks, dynamic graph diffusion convolution, and gated recurrent units to unravel spatio-temporal dependencies.
- **GTCNN** (Sabbaqi & Isufi, 2023) employs product graphs to transform dynamic network data into static data and utilizes the convolution mechanism to capture spatio-temporal dependencies.
- **LEISN** (Lai & Chen, 2024) incorporates a long-term dependency to capture hidden states from multiple previous time steps using graph convolution.
- **MTESformer** (Zhao et al., 2024b) enhances long-term traffic flow prediction by capturing complex spatial-temporal dependencies using multi-scale convolution and transformer techniques.
- **LGSTGCN** (Zong et al., 2024) captures global and local spatial dependencies using a graph attention residual network and a graph convolution network combined with a gated recurrent unit and extracts long-term temporal patterns with a transformer layer.

4.2. Datasets

To validate the performance of our model, we conducted experiments on two public real-world traffic speed datasets, METR-LA and PEMS-BAY. Indeed, these two

datasets are currently the most widely used forecasting datasets, gathering highway traffic data using fixed detectors, including probes, cameras, and loop detectors.

METR-LA has 207 freeway loop detectors that monitor traffic speed information for four months in Los Angeles. PEMS-BAY has 325 loop detectors that monitor traffic speed information for six months in the Bay Area of California. Both databases collect traffic data every 5 minutes. We use the following data split: 70% as a training set, 10% as a validation set, and 20% as a test set.

To ensure data quality and temporal continuity, missing values were imputed using linear interpolation, following the approach of Mallick et al. (2020), which estimates missing values based on adjacent time steps. Outliers were handled using the Isolation Forest algorithm (Liu et al., 2008), a robust unsupervised method for anomaly detection that isolates data points deviating significantly from the majority. The identified anomalous values were removed and subsequently replaced using linear interpolation. To ensure consistency across input features, we applied Z-score normalization, a standardization technique that transforms each value as shown in Equation 12.

$$z = \frac{x - \mu}{\sigma} \tag{12}$$

where x is the original value, μ is the mean, and σ is the standard deviation of the feature values. This transformation aligns the data distributions across sensors and improves training stability.

We considered the detectors as nodes in the graph, and we used the predefined adjacency matrix in Li et al. (2018), which is generated based on the distance of the pairwise road network between detectors.

4.3. Experimental Settings

We conducted the tests in a Linux virtual environment configured with a single core of an Intel(R) Xeon(R) @ 2.00GHz CPU and an NVIDIA Tesla T4 GPU. The model inputs are divided into three categories: weekly, daily, and hourly. We use these inputs to predict the next hour in the future (12 time steps). The weekly and daily inputs contain three channels of features: traffic speed, time of day, and day of the week,

while the hourly inputs contain traffic speed only. The outputs have a single channel, which is the traffic speed.

The weekly temporal encoder produces outputs of length 12 with 16 channels, while the daily temporal encoder produces outputs of length 12 with 8 channels. The spatial modules produce outputs of length 12 with 8 channels for the local spatial module and 16 for the global spatial module. We allocate more feature channels to the weekly temporal encoder as well as to the global spatial module than to the daily temporal encoder and the local spatial module. This design choice is typically based on the assumption that long-term dependencies (weekly module) and the full traffic network dependencies (global module) require a richer representation due to their broader scale, while short-term dependencies (daily/local modules) can be captured with fewer channels.

In order to set the dimensionality of the heads in the temporal decoder, we were inspired by the GMAN model (Zheng et al., 2020b), where the authors tuned the parameters on the validation set and achieved the best performance using 8 attention heads in the attention blocks, each with a dimensionality of 8 channels. Building on this, and to process the features extracted by the spatio-temporal encoder, we set the dimensionality of each head in the temporal decoder to 8. We allocate one attention head (8 channels) for short-range features (local spatial and daily features) and two attention heads (16 channels) for long-range features (global spatial and weekly features) to effectively capture broader dependencies. This results in a total of 6 attention heads, ensuring a balanced trade-off between expressiveness and computational efficiency.

We set the Chebyshev filter size K to 5, and we set the number of heads in the global spatial module to 4. We employ a single Chebyshev layer to capture local spatial dependencies, along with one self-attention layer for the global spatial module and another for the temporal encoder module. We conduct an experimental study on various model configurations in Sub-section 4.5 to further analyze the impact of these design choices.

During training, the batch size was set to 8, and we used the Adam optimizer with a learning rate of 0.005. We set the maximum number of epochs to 100. For evaluation, we used Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and

Root Mean Square Error (RMSE) metrics, defined as follows:

$$\text{MAE} = \frac{1}{M}\sum_{i=1}^{M}|y_i - \hat{y}_i| \tag{13}$$

$$\text{MAPE} = \frac{1}{M}\sum_{i=1}^{M}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\ (\%) \tag{14}$$

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(y_i - \hat{y}_i)^2} \tag{15}$$

Where M represents the number of samples in the testset. y_i denotes the ground truth values and $\hat{y}_i$ represents the predicted values.

4.4. Experimental Results

Table 4 compares the results obtained by our model, as well as results reported by the baseline models mentioned earlier, on the PEMS-Bay and METR-LA datasets across three time horizons: Horizon 3 (15 minutes), Horizon 6 (30 minutes), and Horizon 12 (1 hour). We considered testing our model for these three horizons as the nearest future for prediction, because of its practicality in many situations requiring intelligent vehicle routing based on the position of the vehicle and the time to reach the position for which the speed is predicted. Examples of such situations are re-routing emergency vehicles like ambulances, intelligent transportation systems, and traveling.

We note that the prediction results are weak for traditional models such as ARIMA and SVR. These models are time-series statistical models that consider only the time factor and have limited ability to model nonlinear and complex traffic data. In contrast, deep learning models like FC-LSTM and WaveNet achieve better results than traditional models due to their ability to model some nonlinear dependencies. However, these models remain insufficient in terms of efficiency because they neglect spatial information.

The recent models applied to traffic forecasting use deep learning techniques to extract temporal features in addition to the spatial features represented by graphs extracted by GNNs. It can be observed that the majority of recent models that incorporate spatial dependencies outperform older models based on machine learning or deep

Table 4: Comparison of LSTGAN Performance and Various Baselines on PEMS-BAY and METR-LA Datasets for Three Horizons (15 minutes, 30 minutes, and 60 minutes).

PEMS-BAY										
Category	Models	Horizon 3 (15 min)			Horizon 6 (30 min)			Horizon 12 (60 min)		
		MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE
Time-series Models	ARIMA (Williams & Hoel, 2003)	1.62	3.50%	3.30	2.33	5.40%	4.76	3.38	8.30%	6.50
(Baselines)	SVR (Wu et al., 2004)	1.85	3.80%	3.59	2.48	5.50%	5.18	3.28	8.00%	7.08
Deep Learning	WaveNet (Oord et al., 2016)	1.39	2.91%	3.01	1.83	4.16%	4.21	2.35	5.87%	5.43
Models	FC-LSTM (Karimzadeh et al., 2021)	2.05	4.80%	4.19	2.20	5.20%	4.55	2.37	5.70%	4.96
Graph-Based Spatio-temporal Models	STGCN (Yu et al., 2018)	1.36	2.90%	2.96	1.81	4.17%	4.27	2.49	5.79%	5.69
	DCRNN (Li et al., 2018)	1.38	2.90%	2.95	1.74	3.90%	3.97	2.07	4.90%	4.74
	ASTGCN (Guo et al., 2019)	1.52	3.22%	3.13	2.01	4.48%	4.27	2.61	6.00%	5.42
	GWaveNet (Wu et al., 2019)	1.30	2.73%	2.74	1.63	3.67%	3.70	1.95	4.63%	4.52
	GGRNN (Ruiz et al., 2020)	1.33	2.83%	2.81	1.68	3.79%	3.94	2.34	5.21%	5.14
	DST-GCNN (Chen et al., 2020b)	-	-	-	-	-	-	-	-	-
	GMAN (Zheng et al., 2020a)	1.34	2.81%	2.82	1.62	3.63%	3.72	1.86	**4.31%**	4.32
	TransGAT (Wang et al., 2022d)	1.31	2.74%	2.78	1.62	3.66%	3.69	1.89	4.52%	4.38
	SDGCN (Li et al., 2023a)	1.35	2.89%	2.88	1.69	3.87%	3.91	1.99	4.69%	4.59
	GTCNN (Sabbaqi & Isufi, 2023)	1.25	2.61%	2.66	1.65	3.82%	3.68	2.27	5.11%	4.99
	LEISN (Lai & Chen, 2024)	1.36	2.86%	2.81	1.68	3.81%	3.74	1.98	4.67%	4.52
	MTESformer (Zhao et al., 2024b)	1.31	2.75%	2.80	1.62	3.62%	3.73	1.87	4.35%	4.38
	LGSTGCN (Zong et al., 2024)	**1.04**	**1.90%**	**1.52**	**1.42**	**2.61%**	**2.15**	2.56	4.98%	4.43
	LSTGAN (Ours)	1.33	2.79%	2.80	1.62	3.66%	3.63	**1.84**	4.35%	**4.18**
METR-LA										
Category	Models	Horizon 3 (15 min)			Horizon 6 (30 min)			Horizon 12 (60 min)		
		MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE
Time-series Models	ARIMA (Williams & Hoel, 2003)	3.99	9.60%	8.21	5.15	12.70%	10.45	6.90	17.40%	13.23
(Baselines)	SVR (Wu et al., 2004)	3.99	9.30%	8.45	5.05	12.10%	10.87	6.72	16.70%	13.76
Deep Learning	WaveNet (Oord et al., 2016)	2.99	8.04%	5.89	3.59	10.25%	7.28	4.45	13.62%	8.93
Models	FC-LSTM (Karimzadeh et al., 2021)	3.44	9.60%	6.30	3.77	10.90%	7.23	4.37	13.20%	8.69
Graph-Based Spatio-temporal Models	STGCN (Yu et al., 2018)	2.88	7.62%	5.74	3.47	9.57%	7.24	4.59	12.70%	9.40
	DCRNN (Li et al., 2018)	2.80	7.23%	5.40	3.18	8.65%	6.41	3.63	10.40%	7.49
	ASTGCN (Guo et al., 2019)	4.86	9.21%	9.27	5.43	10.13%	10.61	6.51	11.64%	12.52
	GWaveNet (Wu et al., 2019)	2.69	6.90%	5.15	3.07	8.37%	6.22	3.53	10.01%	7.37
	GGRNN (Ruiz et al., 2020)	2.73	7.12%	5.44	3.31	8.97%	6.63	3.88	10.59%	8.14
	DST-GCNN (Chen et al., 2020b)	2.68	7.20%	5.35	3.01	8.50%	6.23	3.41	10.30%	7.47
	GMAN (Zheng et al., 2020a)	2.77	7.25%	5.48	3.07	8.35%	6.34	3.40	9.72%	7.21
	TransGAT (Wang et al., 2022d)	2.71	7.69%	5.20	3.15	8.70%	6.16	3.48	9.80%	7.21
	SDGCN (Li et al., 2023a)	2.76	7.16%	5.38	3.15	8.53%	6.46	3.61	9.87%	7.43
	GTCNN (Sabbaqi & Isufi, 2023)	2.68	6.85%	5.17	3.02	8.30%	6.20	3.55	10.21%	7.35
	LEISN (Lai & Chen, 2024)	2.77	7.18%	5.29	3.13	8.48%	6.33	3.52	9.97%	7.40
	MTESformer (Zhao et al., 2024b)	2.73	7.10%	5.34	3.03	8.30%	6.22	3.37	9.62%	7.14
	LGSTGCN (Zong et al., 2024)	**1.85**	**3.87%**	**3.29**	**2.55**	**7.33%**	**4.69**	4.31	10.89%	7.26
	LSTGAN (Ours)	2.50	6.37%	4.98	2.90	7.87%	5.95	**3.32**	**9.42%**	**6.87**

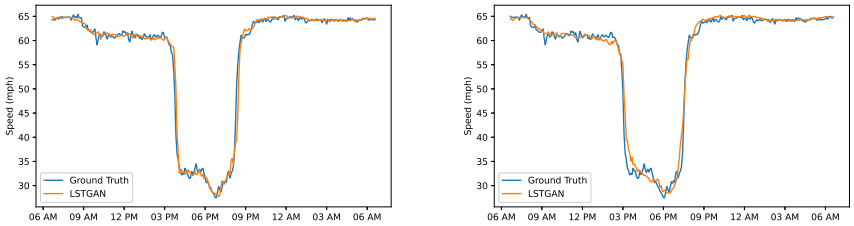
* Results in **bold** indicate the best performance, while those with underlining represent the second-best performance.

learning without spatial dependencies, with the exception of the ASTGCN, which may lack the ability to explore long-term spatio-temporal dependencies. This indicates that spatial information is essential to achieve high accuracy in traffic forecasting models.

The performance analysis of our LSTGAN model, compared to other models, demonstrates its superiority at the long-term horizon (12), which is considered a challenging forecasting horizon and typically more difficult to predict than shorter horizons. However, we observe that the LGSTGCN model achieves slightly better performance at shorter horizons (3 and 6). This can be attributed to its architectural design, which first captures global spatial dependencies to enrich spatial semantic representation, and then integrates real-time local spatial-temporal dependencies, with a strong focus on the dynamic influence of neighboring roads at the current time step. This flow from global to local context tends to prioritize immediate spatio-temporal signals, which, while effective for short-term prediction, may attenuate the influence of earlier captured global information. In contrast, LSTGAN is explicitly designed to emphasize long-range spatio-temporal patterns derived from weekly and daily data, alongside diverse local and global spatial dependencies, organized in a parallel modular structure. This architecture enables LSTGAN to better capture broader temporal regularities, offering a distinct advantage in long-term forecasting tasks.

Figure 7 visualizes the predicted values alongside the ground truth values for Horizon 3 (15 minutes) and Horizon 12 (1 hour) on sensor 145 in the PEMS-BAY dataset. It demonstrates the ability of our model to accurately capture rapid traffic trends and fluctuations in short-term forecasting while maintaining this accuracy with minimal delay in detecting changing traffic patterns, even for long-term predictions up to one hour ahead.

Discussion. The previous results demonstrate LSTGAN's ability to capture long-term historical spatial and temporal dependencies. By utilizing long-term weekly information, the model learns recurring traffic trends at similar times of the same day and hour from the previous week. This allows it to capture patterns that are stable over time but may not be apparent from short-term data solely. These weekly patterns provide an underlying context that reflects consistent behavioral trends, such as weekday



(a) Short-term forecasting (15 minutes).

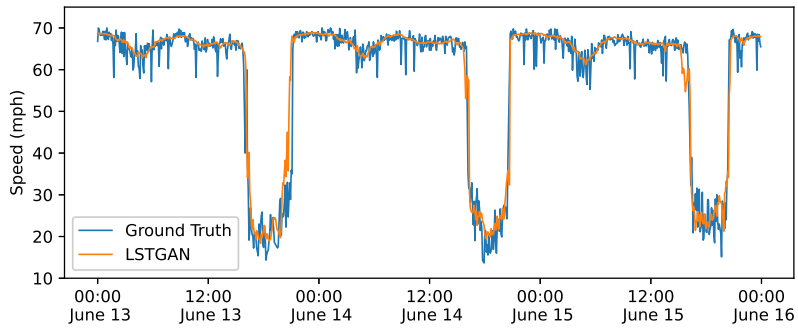
(b) Long-term forecasting (60 minutes).

Figure 7: Visualization of forecasted and ground truth values for Horizon 3 (15 minutes) and Horizon 12 (1 hour) on sensor 145 in the PEMS-BAY dataset.

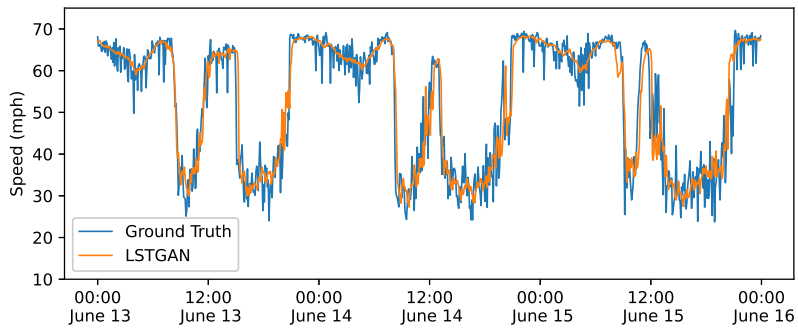
commuting peaks versus variations on weekends, which enhances the model's understanding of traffic flow. Furthermore, traffic data from the last 24 hours is crucial for traffic forecasts. It captures recent traffic trends, patterns, and any perturbations or events that may directly affect traffic conditions within the following hour. This information enables the model to adapt to daily cycles, such as morning and evening peak hours. Moreover, LSTGAN captures spatial information from the last 60 minutes for neighboring roads through graph convolution, providing detailed insights that are specific to the environment, thus enabling the model to respond quickly to changes and effectively incorporate the impacts of local events. Besides, our model captures spatial information from all roads in the network by leveraging spatial attention across all sensors, which offers a comprehensive context for the entire road network, helping in identifying broader traffic patterns and in recognizing interconnections between roads. This forms a latent spatio-temporal representation useful for effective forecasting, as it allows the model to produce more accurate and nuanced traffic predictions.

Prediction Stability Across Time Periods. To assess the stability of our model across varying traffic conditions, we include a visual comparison of predicted and actual traffic speeds during different times of day. Figure 8 presents the prediction results of the proposed LSTGAN model for two sensors (Sensor No. 97 and Sensor No. 141) on the METR-LA dataset over a continuous three-day period. The model consistently captures traffic speed trends throughout the day, including both peak and off-peak pe-

riods. For Sensor No. 97 (Figure 8a), which exhibits a regular daily pattern with a distinct evening congestion peak, the model demonstrates strong alignment with the ground truth. It maintains stability during off-peak hours, accurately preserving high-speed flow, and effectively tracks the onset and dissipation of congestion during peak hours. Sensor No. 141 (Figure 8b) presents a more complex pattern, including multiple congestion events and varying speed fluctuations. Despite this variability, the model accurately predicts both the magnitude and timing of changes in traffic speed, indicating strong generalization across non-uniform patterns. Notably, the model smooths high-frequency noise while remaining responsive to rapid transitions, confirming its robustness across a wide range of traffic dynamics.



(a) Prediction results on Sensor No. 97.



(b) Prediction results on Sensor No. 141.

Figure 8: Prediction results on METR-LA for a three-day period.

4.5. Model Hyperparameter Analysis

We conduct an experimental study on key hyperparameters to evaluate their impact on the performance of the model. Specifically, we evaluate the impact of the size of the Chebyshev filter size (K) that controls the receptive field of the graph convolution, the number of layers of the graph convolutional network (GCN), and the number of self-attention layers in the global spatial encoder and the temporal decoder. By systematically varying these parameters, we evaluate their impact on the performance of the model to determine the optimal configuration. Figure 9 presents the experimental results obtained by varying (a) the Chebyshev filter size K, (b) the number of GCN layers, and (c) the number of self-attention layers, and their impact on the average MAE, MAPE, and RMSE in the LSTGAN model on the METR-LA dataset.

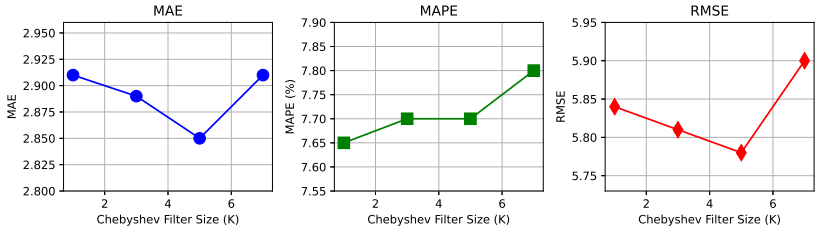
The experimental results in Figure 9a illustrate that increasing the filter size K enhances the model's ability to capture spatial dependencies, up to an optimal value of $K = 5$. Beyond this point, performance starts to degrade. This behavior can be explained by the over-smoothing phenomenon in graph convolutional networks. As the order of the Chebyshev polynomials increases, node features become overly similar, which blurs important local patterns. Additionally, higher-order filters may amplify high-frequency noise in the graph signal.

The experimental results in Figure 9b show that a single GCN layer provides the best performance across all metrics. As the depth increases, the model's performance begins to degrade slightly.

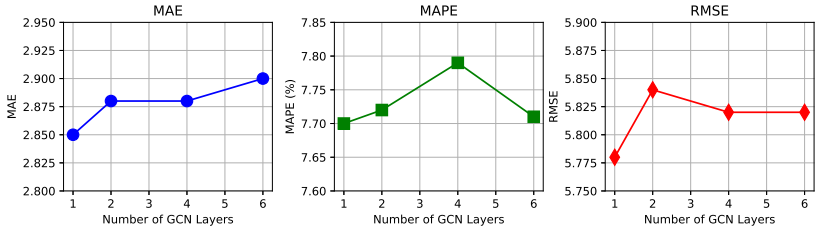
In Figure 9c, the results show that a single self-attention layer in the Temporal Decoder achieves the best performance across all metrics. As the depth increases, the model's performance slightly degrades. Similarly, Figure 9d, which represents the variation in attention layers in the Global Spatial Encoder, also shows that a single layer yields the best performance.

4.6. Feature Fusion Strategies

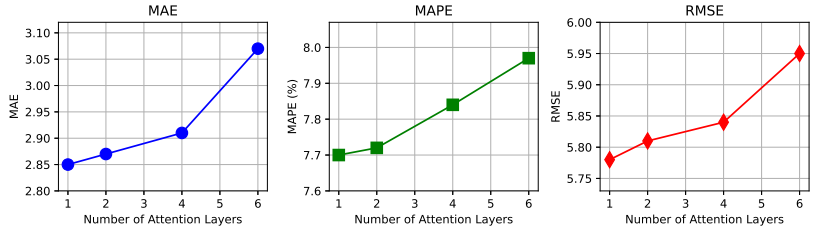
Our proposed model fuses historical features with local and global spatial features in parallel, enabling it to capture multi-scale dependencies simultaneously. This par-



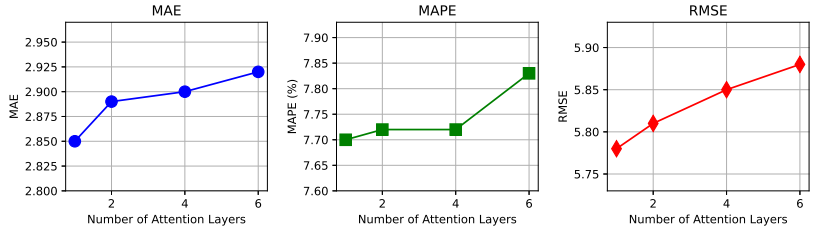
(a) Varying the size of the Chebyshev filter size (K).



(b) Varying the number of GCN layers.



(c) Varying the number of Temporal Decoder attention layers.



(d) Varying the number of the Global Spatial Encoder attention layers.

Figure 9: Performance analysis of varying model hyperparameters on the METR-LA dataset.

allel fusion enriches feature representations and allows the attention-based decoder to effectively identify the most impactful information for forecasting.

To empirically validate this design choice, we conduct an experiment comparing our parallel fusion strategy with the sequential fusion strategy, where features are progressively integrated. Specifically, we evaluate two sequential fusion strategies:

- **Local-to-Global** Fusion: we first extract local spatial features from historical data. These features are then used to extract global spatial features, incorporating broader spatial dependencies.

- **Global-to-Local** Fusion: we begin by extracting global spatial features to establish a high-level contextual representation, which we then refine by extracting local spatial features for finer spatial details.

Finally, the temporal decoder processes the fused features to generate the final predictions.

Table 5: Performance Comparison of Parallel and Sequential Fusion Strategies on the METR-LA Dataset.

Fusion Strategy	Horizon 3 (15 min)			Horizon 6 (30 min)			Horizon 12 (60 min)		
	MAE	**MAPE**	**RMSE**	**MAE**	**MAPE**	**RMSE**	**MAE**	**MAPE**	**RMSE**
Local-to-Global	2.58	6.62%	5.04	3.00	8.10%	6.02	3.50	10.00%	7.06
Global-to-local	2.55	6.55%	5.02	2.93	7.99%	5.97	3.38	9.67%	6.97
Parallel	**2.50**	**6.37%**	**4.98**	**2.90**	**7.87%**	**5.95**	**3.32**	**9.42%**	**6.87**

The results in the table 5 demonstrate that the Parallel Fusion strategy consistently outperforms both Local-to-Global and Global-to-Local fusion approaches across all forecasting horizons, further validating our design choice.

Among the sequential strategies, Global-to-Local Fusion performs slightly better than Local-to-Global Fusion, suggesting that capturing global spatial dependencies first provides a more informative representation before refining it with local details. However, both sequential strategies lag behind the Parallel Fusion approach, highlighting the advantage of simultaneously integrating multi-scale dependencies rather than processing them sequentially.

4.7. Ablation Study

To evaluate the effectiveness of different parts of our model, we conduct two ablation studies. The first focuses on evaluating the effectiveness of the attention mechanism in our model, while the second focuses on evaluating the impact of other components of the model.

4.7.1. Impact of Attention Mechanism

To evaluate the effectiveness of the self-attention mechanism in our model, we create a new variant of the **LSTGAN** named **LSTGAN w/o AT**, by completely removing the attention mechanism from the model (removing the Global Spatial Encoder and replacing the Temporal Decoder in the forecasting unit with an LSTM Encoder-Decoder). We then performed ablation experiments on the PEMS-BAY and METR-LA datasets to assess the impact of these changes and the contribution of the self-attention mechanism to overall model performance.

The Long Short Term Memory (LSTM) encoder-decoder that we used is a type of Recurrent Neural Network (RNN) designed to capture long-term temporal dependencies in sequential data. The encoder compresses the input sequence into a context vector, which the decoder then uses to generate the output sequence. This architecture is especially effective for tasks with complex, long-range dependencies.

The encoder and decoder networks consist of LSTM layers. The encoder processes historical traffic data as a sequence of T time steps $X = \left\{x^1, x^2.., x^T\right\}$ and generates hidden states $h^t = f(h^{t-1}, x^t)$ using a non-linear activation function f. The last hidden state h^T is repeated to match the desired output length T', forming a sequence $H = \left\{h^{T_1}, h^{T_2}.., h^{T_{T'}}\right\}$ to be used as input to the decoder. The decoder then produces hidden states $h^{t'} = g(h^{t'-1}, h^{T_{t'}})$ with a non-linear activation function g. Finally, these states are processed through a linear layer to obtain the final predictions $\mathcal{F} = \left\{f^1, f^2.., f^{T'}\right\}$.

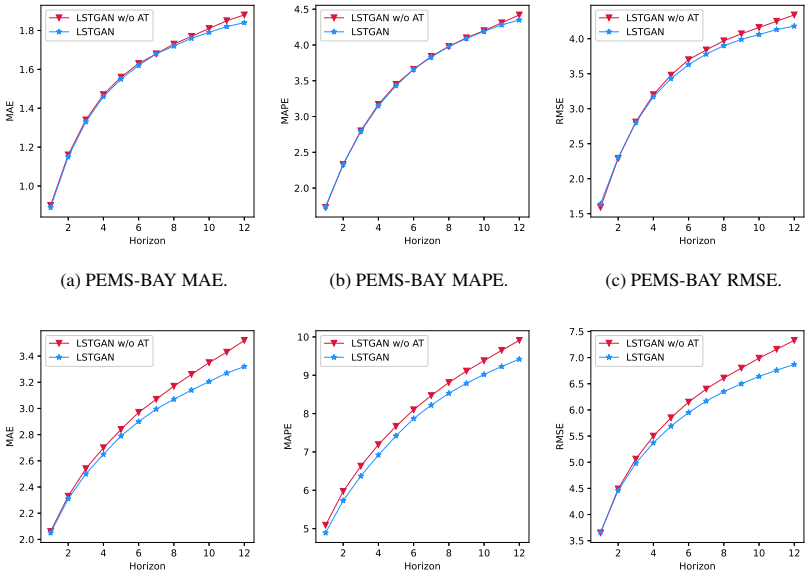
The results of our experiments on the changed model are shown in Table 6. As shown in Figure 10, we observed that the original version of LSTGAN performed almost equally to LSTGAN w/o AT in the short-term horizons. However, we observe a significant decrease in the ability to make long-term predictions (starting from Horizon 9 and beyond) due to the absence of information extracted by self-attention from distant

Table 6: Ablation Experiment for LSTGAN and LSTGAN without Attention Mechanism on the PEMS-BAY and METR-LA Datasets.

PEMS-BAY									
Models	Horizon 3 (15 min)			Horizon 6 (30 min)			Horizon 12 (60 min)		
	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE
LSTGAN w/o AT	1.34	2.80%	2.83	1.63	**3.66%**	3.70	1.88	4.42%	4.34
LSTGAN	**1.33**	**2.79%**	**2.80**	**1.62**	**3.66%**	**3.63**	**1.84**	**4.35%**	**4.18**

METR-LA									
Models	Horizon 3 (15 min)			Horizon 6 (30 min)			Horizon 12 (60 min)		
	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE
LSTGAN w/o AT	2.54	6.63%	5.06	2.97	8.10%	6.15	3.52	9.91%	7.33
LSTGAN	**2.50**	**6.37%**	**4.98**	**2.90**	**7.87%**	**5.95**	**3.32**	**9.42%**	**6.87**

sensors within the road network and the error accumulation, gradient explosion, and vanishing in RNNs. Thus, it is clear that the attention mechanism is effective in both capturing spatial data and processing temporal data.



(a) PEMS-BAY MAE. (b) PEMS-BAY MAPE. (c) PEMS-BAY RMSE.

(d) METR-LA MAE. (e) METR-LA MAPE. (f) METR-LA RMSE.

Figure 10: Comparison of MAE, MAPE, and RMSE values across the 12 horizons for the ablation study on the PEMS-BAY and METR-LA testset.

4.7.2. Impact of Model Components

The previous ablation study revealed a performance degradation on both the PEMS-BAY and METR-LA datasets. Therefore, we conducted additional ablation experiments on the LSTGAN model using the METR-LA dataset to assess the impact of its key components on the overall performance. The different variants of the LSTGAN model are named as follows:

- **LSTGAN w/o TE:** Excluding time-related information (time of day and day of the week) by removing the time encoding unit.
- **LSTGAN w/o DE:** Removing the daily Long-Term Temporal Encoder from the Spatio-temporal Unit.
- **LSTGAN w/o WE:** Removing the weekly Long-Term Temporal Encoder from the Spatio-temporal Unit.
- **LSTGAN w/o D&W:** Removing the daily and weekly Long-Term Temporal Encoder from the Spatio-temporal Unit.
- **LSTGAN w/o GCN:** Removing the Local Spatial Encoder from the Spatio-temporal Unit.
- **LSTGAN w/o SA:** Removing the Global Spatial Encoder from the Spatio-temporal Unit.

Table 7: Ablation Experiment for LSTGAN and Its Variants on the METR-LA Dataset.

Models	Horizon 3 (15 min)			Horizon 6 (30 min)			Horizon 12 (60 min)		
	MAE	**MAPE**	**RMSE**	**MAE**	**MAPE**	**RMSE**	**MAE**	**MAPE**	**RMSE**
LSTGAN w/o TE	2.54	6.55%	5.03	2.95	8.02%	5.98	3.40	9.90%	6.88
LSTGAN w/o DE	2.58	6.71%	5.07	3.00	8.29%	6.03	3.49	10.11%	7.06
LSTGAN w/o WE	2.60	6.66%	5.12	3.04	8.34%	6.17	3.59	10.45%	7.29
LSTGAN w/o D&W	2.80	7.37%	5.42	3.34	9.36%	6.71	4.14	12.49%	8.32
LSTGAN w/o GCN	2.55	6.48%	5.07	2.94	7.92%	6.00	3.39	9.56%	6.96
LSTGAN w/o SA	2.51	6.39%	5.03	2.92	7.93%	6.07	3.34	9.50%	7.04
LSTGAN	**2.50**	**6.37%**	**4.98**	**2.90**	**7.87%**	**5.95**	**3.32**	**9.42%**	**6.87**

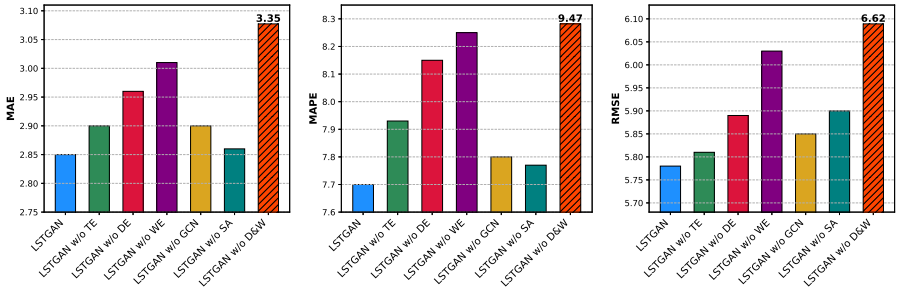
The results of this ablation experiment are presented in Table 7, while Figure 11 provides a comparison of the average metrics across all **LSTGAN** variations. These

results highlight the contribution of each component to the overall model performance, with some components having a more pronounced effect than others. When removing the time coding (**LSTGAN w/o TE**) the performance decreases, demonstrating the importance of time-related contextual information.

The largest drop in performance occurs when both the daily and weekly long-term encoders are removed (**LSTGAN w/o D&W**), which indicates that capturing long-term temporal dependencies is critical for accurate prediction. Removing the daily or weekly encoder separately also leads to a decrease in performance, although to a lesser degree, confirming their complementary roles in modeling periodic trends.

Removing the local spatial encoder (**LSTGAN w/o GCN**) or the global spatial encoder (**LSTGAN w/o SA**) shows a decline in the model's performance. This indicates that the LSTGAN model takes benefit of the information from neighboring and distant sensors within the road network, and highlights the importance of both local and global spatial dependencies in improving prediction accuracy.

The full **LSTGAN** model achieves the best results, proving that all spatial and temporal modules contribute to the overall performance. The results confirm that long-term temporal dependencies (daily and weekly) are the most impactful, while spatial dependencies further improve predictions.



(a) MAE (b) MAPE (c) RMSE

Figure 11: Comparison of Average MAE, MAPE, and RMSE Across LSTGAN Variants for the METR-LA testset.

5. Conclusion

In this paper, we propose LSTGAN, a Long-Term Spatio-Temporal Graph Attention Network designed to analyze long-term historical data for traffic forecasting. Our model integrates temporal and spatial dependencies using temporal encoders that extract and summarize long-term temporal information using convolutional layers, along with spatial encoders that capture spatial features using the Spatial Attention module and a GCN. We employed a temporal decoder to perform predictions based on the extracted information. Compared to previous works on METR-LA and PEMS-BAY datasets, LSTGAN demonstrates superior performance in most metrics for long-term scenarios. This superiority is owing to our model's ability to effectively identify trends and patterns in long-term temporal dependencies, as well as to its ability to capture spatial correlations across the road network, which contribute to improved prediction accuracy. This point requires to be deeply explored in further work to understand the dynamic decay pattern of prediction errors as the forecasting horizon increases. Specifically, the pattern of error variation with forecasting duration (e.g., linear growth, exponential saturation, or segmental stability) will be investigated. This will allow us to understand LSTGAN's anti-decay capability and facilitate a clear comparison of its stability with baseline models in long-term forecasting. In addition, we will investigate the robustness of LSTGAN in scenarios with incomplete data and also its generalization on diverse urban layouts (e.g., sparse sensor networks). We also plan to employ high-order GNNs (Giraldo et al., 2024, 2023) in order to enhance the representation of long-term temporal dependencies by capturing spatial correlations within historical data to better describe traffic trends and patterns and then to evaluate it in long-term scenarios with unseen spatial/temporal patterns. Additionally, we will perform a time series analysis to detect periodic behaviors, such as seasonality, and integrate this information into the model as a new attention mechanism. Furthermore, we aim to investigate the incorporation of external factors, such as weather conditions and public events, to improve the model's robustness and forecasting accuracy. We will also investigate a real-time implementation (Jeziorek et al., 2023, 2024a,b) to provide a model that can handle dynamic spatio-temporal complexities while keeping computational efficiency.

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Appendix A. Glossary of Acronyms

Table A.8: Glossary of Acronyms.

Acronym	Definition
AE	AutoEncoder
ARIMA	Auto-Regressive and Integrated Moving Average
ASTGCN	Attention based Spatial-Temporal Graph Convolutional Networks
CNN	Convolutional Neural Network
DCRNN	Diffusion Convolutional Recurrent Neural Network
DST-GCNN	Dynamic Spatio-Temporal Graph-based Convolutional Neural Networks
FC-LSTM	Fully-Connected Long Short Term Memory
GCN	Graph Convolutional Network
GGRNN	Gated Graph Recurrent Neural Networks
GLLM	Graph Large Language Models
GMAN	Graph Multi-Attention Network
GNN	Graph Neural Network
GRU	Gated recurrent unit
GTCNN	Graph-Time Convolutional Neural Networks
GTN	Graph Transformer Network
LEISN	Long Explicit–Implicit Spatio-temporal Network
LGSTGCN	Local-Global Spatial-Temporal Graph Convolutional Network
LSTGAN	Long-Term Spatio-Temporal Graph Attention Network
LSTM	Long Short Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
METR-LA	Los Angeles Metropolitan Traffic Dataset
MLP	MultiLayer Perceptron
MTESformer	Multi-scale Temporal and Enhance Spatial Transformer
PEMS-BAY	California Bay Area Traffic Dataset
RMSE	Root Mean Square Error

Acronym	Definition
RNN	Recurrent Neural Networks
SDGCN	Spatial Dynamic Graph Convolutional Network
STGCN	Spatio-Temporal Graph Convolutional Network
STGNN	Spatial-Temporal Graph Neural Network
SVR	Support Vector Regression
TransGAT	Dynamic Graph Attention Residual Networks